Sepehr Hajebi – CV

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Contact

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Appointments

- **University of Toronto**, Department of Mathematics — 2026 – present
  Postdoctoral Fellow holding a Canada Postdoctoral Research Award
- **University of Waterloo**, Combinatorics and Optimization — 2024 – 2026
  Postdoctoral Scholar
- **Princeton University**, Department of Mathematics — 2024
  Junior Faculty (Instructor)
  Not taken up due to US entry restriction against all Iranian nationals

Education

- **University of Waterloo**, PhD in Combinatorics and Optimization — 2020 – 2024
  Thesis: Foreshadowing the Grid Theorem for Induced Subgraphs
  Advisor: Sophie Spirkl
- **Isfahan University of Technology**, BSc and MSc in Mathematics — 2013 – 2019

Awards

- **Canada Postdoctoal Research Award**, 2026 (140,000 CAD)
  Natural Sciences and Engineering Research Council of Canada (NSERC)
- **Graduate Research Excellence Award**, 2025 (2500 CAD)
  University of Waterloo
- **Mathematics Doctoral Prize: 1st place**, 2025 (1500 CAD)
  University of Waterloo
- **Governor General's Gold Medal**, Finalist Designation, 2025
  University of Waterloo
- **Sinclair Graduate Scholarship**, 2023-2024 (4000 CAD)
  University of Waterloo
- **Outstanding TA Award**, 2022
  University of Waterloo
- **Math competitions:** *International Mathematics Competition for University Students* (Bronze Medals – 2015 and 2016), *Korean Mathematical Contest* (Bronze Medal – 2016), *Iranian Mathematical Competition* (Silver and Bronze Medals – 2015 and 2016)
- **Elite Student Recognition**, 2016 and 2018
  Isfahan University of Technology

Publications

▶ Journal papers

[1] *Bull-free graphs and χ-boundedness*
Innovations in Graph Theory 3 (2025), 247–253.

[2] *Induced subgraphs and tree decompositions XIX. Thetas and forests*
SIAM Journal on Discrete Mathematics 40 (2026), 1000–1009.
with M. Chudnovsky, J. Codsi and S. Spirkl.

[3] *Tree independence number III. Thetas, prisms and stars*
Journal of Graph Theory 106 (2024), 923–943.
with M. Chudnovsky and N. Trotignon.

[4] *Suns in triangle-free graphs of large chromatic number*
Electronic Journal of Combinatorics 33 (2026)
with S. Spirkl.

[5] *Induced subgraphs and tree decompositions XVI. Complete bipartite induced minors*
Journal of Combinatorial Theory, Series B 176 (2026), 287–318.
with M. Chudnovsky and S. Spirkl.

[6] *Tree independence number II. Three path configurations*
Journal of Combinatorial Theory, Series B 176 (2026), 74–96.
with M. Chudnovsky, D. Lokshtanov and S. Spirkl.

[7] *Chordal graphs, even-hole-free graphs and sparse obstructions to bounded treewidth*
Journal of Graph Theory (2025), 1–15.

[8] *Induced subgraphs and tree decompositions IX. Grid theorem for perforated graphs*
Advances in Combinatorics 2025:3, 40pp.
with B. Alecu, M. Chudnovsky and S. Spirkl.

[9] *Induced subgraphs and tree decompositions XII. Grid theorem for pinched graphs*
Innovations in Graph Theory 2 (2025), 1–23.
with B. Alecu, M. Chudnovsky and S. Spirkl.

[10] *Induced subgraphs and tree decompositions XIV. Non-adjacent neighbors in a hole*
European Journal of Combinatorics 124 (2025), 104–074.
with M. Chudnovsky and S. Spirkl.

[11] *Induced subdivisions with pinned branch vertices*
European Journal of Combinatorics 124 (2025), 104–072.

[12] *Induced subgraphs and tree decompositions VI. Graphs with 2-cutsets*
Discrete Mathematics 348 (2025), 114–195.
with T. Abrishami, M. Chudnovsky and S. Spirkl.

[13] *Induced subgraphs and tree decompositions XIII. Basic obstruction in $\mathcal{H}$-free graphs for finite $\mathcal{H}$*
Advances in Combinatorics 2024:6, 30pp.
with B. Alecu, M. Chudnovsky and S. Spirkl.

[14] *List-k-Coloring H-free graphs for all $k \geq 4$*
Combinatorica 44 (2024). 1063–1068.
with M. Chudnovsky and S. Spirkl.

[15] *Induced subgraphs and tree decompositions VIII: Excluding a forest in (theta, prism)-free graphs*
Combinatorica 44 (2024), 921–948.
with T. Abrishami, B. Alecu, M. Chudnovsky and S. Spirkl.

[16] *Tree independence number I. (Even hole, diamond, pyramid)-free graphs*
Journal of Graph Theory 106 (2024), 923–943.
with T. Abrishami, B. Alecu, M. Chudnovsky. S. Spirkl and K. Vŭsković.

[17] *List-3-Coloring ordered graphs with a forbidden induced subgraphs*
SIAM Journal on Discrete Mathematics 38 (2024), 1158–1190.
with Y. Li and S. Spirkl.

[18] *Hitting all maximum stable sets in P_5-free graphs*
Journal of Combinatorial Theory, Series B 165 (2024), 142–163.
with Y. Li and S. Spirkl.

[19] *Induced subgraphs and tree decompositions VII. Basic obstructions in H-free graphs*
Journal of Combinatorial Theory, Series B 164 (2024), 443–472.
with T. Abrishami, B. Alecu, M. Chudnovsky and S. Spirkl.

[20] *Induced subgraphs and tree decompositions II. Toward walls and their line graphs in graphs of bounded degree*
Journal of Combinatorial Theory, Series B 164 (2024), 371–403.
with T. Abrishami, M. Chudnovsky, C. Dibek, P. Rzążewski, S. Spirkl and K. Vŭsković.

[21] *Induced subgraphs and tree decompositions V. One neighbor in a hole*
Journal of Graph Theory 105 (2023), 542–561.
with T. Abrishami, B. Alecu, M. Chudnovsky. S. Spirkl and K. Vŭsković.

[22] *Induced subgraphs and tree decompositions IV. (Even hole, diamond, pyramid)-free graphs*
Electronic Journal of Combinatorics 30 (2023)
with T. Abrishami, M. Chudnovsky and S. Spirkl.

[23] *Induced subgraphs and tree decompositions III. Three-path-configurations and logarithmic treewidth*
Advances in Combinatorics 2022: 6, 29 pp.
with T. Abrishami, M. Chudnovsky and S. Spirkl.

[24] *Complexity dichotomy for List-5-Coloring with a forbidden induced subgraph*
SIAM Journal on Discrete Mathematics 36 (2022), 2004–2027.
with Y. Li and S. Spirkl.

[25] *Minimal induced subgraphs of two classes of 2-connected non-Hamiltonian graphs*
Discrete Mathematics 345 (2022).
with J. Cheriyan, Z. Qu and S. Spirkl.

[26] *Edge clique cover of claw-free graphs*
Journal of Graph Theory 90 (2019), 311–405.
with R. Javadi.

▶ Preprints

[27] *Forcing monochromatic induced subgraphs*
`arXiv:2606.24695`
with S. Spirkl.

[28] *Asymmetric induced saturation*
`2606.24763`
with X. Fan, Sahab Hajebi and S. Spirkl.

[29] *Tree-alpha and excluding finitely many graphs*
`arXiv:2605.01223`
with S. Spirkl.

[30] *Polynomial bounds for pathwidth*
`arXiv:2510.19120`.

[31] *A simple layered-wheel-like construction*
`arXiv:2410.16495`
with M. Chudnovsky, D. Fischer, S. Spirkl and B. Walczak.

[32] *Halfway to induced saturation for even cycles*
`arXiv:2505.24100`
with X. Fan, Sahab Hajebi and S. Spirkl.

[33] *Induced subgraphs and tree decompositions XVIII. Obstructions to bounded pathwidth*
`arXiv:2412.17756`.
with M. Chudnovsky and S. Spirkl.

[34] *Induced subgraphs and tree decompositions XVII. Anticomplete sets of large treewidth*
`arXiv:2411.11842`.
with M. Chudnovsky and S. Spirkl.

[35] *Tree independence number IV. Even-hole-free graphs*
`arXiv:2407.08927`
• Conference version at **SODA (2025)**, 4444–4461.
with M. Chudnovsky, P. Gartland, D. Lokshtanov and S. Spirkl.

[36] M. Chudnovsky, P. Gartland, S. Hajebi, D. Lokshtanov and S. Spirkl
Induced subgraphs and tree decompositions XV. Even-hole-free graphs have logarithmic treewidth
with M. Chudnovsky, D. Lokshtanov and S. Spirkl.

[37] *Induced subgraphs and tree decompositions XI. Local structure in even-hole-free graphs of large treewidth*
`arXiv:2309.04390`
with B. Alecu, M. Chudnovsky and S. Spirkl.

[38] *Induced subgraphs and tree decompositions X. Towards logarithmic treewidth for even-hole-free graphs*
`arXiv:2307.13684`
with T. Abrishami, B. Alecu, M. Chudnovsky and S. Spirkl.

Talks

- *Polynomial pathwidth* (2026)
  Barbados Graph Theory Workshop, Bellairs Research Institute in Holetown, Barbados.
- *Complete bipartite induced minors (and treewidth)* (2025)
  Willaim Tutte Colloquium, University of Waterloo.
- *Local and global structure of non-basic obstructions to bounded treewidth* (2025)
  CanaDAM (invited at the Structural Graph Theory Minisymposium), University of Ottawa.
- *"Underview" of excluded induced subgraphs for bounded pathwidth* (2025)
  Barbados Graph Theory Workshop, Bellairs Research Institute, Holetown, Barbados.
- *The pathwidth theorem for induced subgraphs* (2025)
  IBS discrete math seminar, Daejeon, South Korea.
- *The pathwidth theorem for induced subgraphs* (2024)
  Graphs and Matroids Seminar, University of Waterloo.
- *Anticomplete sets of large treewidth* (2024)
  BIRS Workshop on New Perspectives in Coloring and Structure, Banff, Alberta.

- *Survey on induced subgraphs of graphs with large treewidth* (2024)
  Barbados Graph Theory Workshop, Bellairs Research Institute in Holetown, Barbados.
- *Treewidth, Erdős-Pósa and induced subgraphs* (2024)
  New York Combinatorics Seminar. [Virtual]
- *Several Gyárfás-Sumner-type results for treewidth* (2023)
  Graphs and Matroids Seminar, University of Waterloo.
- *Hitting all maximum stable sets in P_5-free graphs* (2023)
  Graphs and Matroids Seminar, University of Waterloo.
- *Forests in even-hole-free graphs of large treewidth* (2022)
  Barbados Graph Theory Workshop, Bellairs Research Institute in Holetown, Barbados.
- *Holes, hubs, and bounded treewidth* (2022)
  IBS Discrete Math Seminar. [Virtual]
- *Bounded treewidth in hereditary graph classes* (2022)
  Graphs and Matroids Seminar, University of Waterloo.
- *Bounded treewidth in hereditary graph classes* (2022)
  Seymour was 70 two years ago, ENS de Lyon, France.

Teaching

- **Teaching at the University of Toronto**
  - MATH 344 Introduction to Combinatorics (Fall 2026)
- **Teaching at the University of Waterloo**
  - CO 380 Mathematical Discovery and Invention (Spring 2026)
  - MATH 135 Algebra for Honours Mathematics (Winter 2026)
  - MATH 116 Calculus I for Engineering (Fall 2025)
- **TA work at the University of Waterloo**
  - CO 342 Graph Theory (Spring 2023)
  - MATH 138 Calculus II for Honors Mathematics (Winter 2023)
  - MATH 600 Mathematical Software (Fall 2022)
  - CO 456 Game Theory (Fall 2022 and 2023)
  - CO 380 Mathematical Discovery and Invention (Spring 2022)
  - MATH 239 Introduction to Combinatorics (Winter 2022)
  - CO 255 Advanced Optimization (Winter 2022)
  - CO 250 Introduction to Optimization (Fall 2021 & 2023, Winter 2023 & 2024, Spring 2024)
  - CO 442/642 Graph Theory (Fall 2021)
  - CO 351 Network-Flow Theory (Spring 2021)
- **TA work at Isfahan University of Technology**
  - Computational Complexity (2019)
  - Elements of Matrices and Linear Algebra (2018)

- Applied Linear Algebra for Engineering (2018)
- Graph Theory (2014, 2016, 2017)
- Elements of Combinatorics (2014, 2016, 2017)

Service

- **Supervision/Mentorship at the University of Waterloo**
  - Graduate students:
    - Xinyue Fan (MSc, 2024 – 2026, joint with Sophie Spirkl)
      Thesis: *On induced saturation in graphs and tournaments*
  - Directed Reading Program (DRP):
    - Spring 2024 project: *Non-constructive methods in combinatorics*
      Mentees: Anna Henderson and Gioia De Leonardis
    - Fall 2023 project: *Introduction to graph minor theory*
      Mentees: Xinyue Fan and Lyncy Li
  - Undergraduate Research Assistant Program (URA):
    - Spring 2023 project: *Maximum transitive set in H-free tournaments*
      Mentee: Yun Xing
- **Organizing**
  - *Themes and trends in structural graph theory* – Organizer
    Workshop at the American Institute of Mathematics (AIM), to be held in 2027
- **Refereeing for journals and conference proceedings:** IMRN, Combinatorica, JCTb, Innov. Graph Theory, Eur. J. Combin., SIDMA, J. Graph Theory, Electron. J. Combin., SODA, EuroComb, Workshop on Graphs.